Who Am I Building This for? How Will They Use It?

Responsible AI: Human-Centered Design

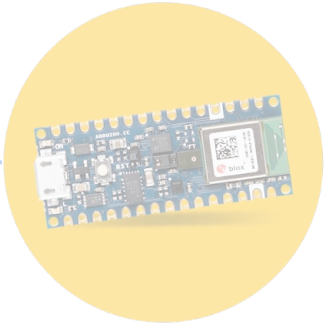


Course 1
Fundamentals of TinyML

- What am I building?
- **Who am I building this for?**
- What are the consequences for the user if it fails?

Course 2
Applications of TinyML

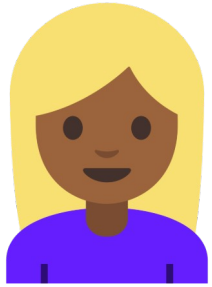
Course 3
Deploying TinyML



Who am I building this for?

Who are the **stakeholders** impacted by the proposed technology?

Direct



aka the “User(s)”

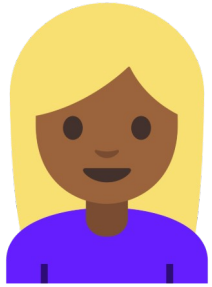
Who am I building this for?

Who are the **stakeholders** impacted by the proposed technology?

Be specific!

- Age
- Gender
- Physical abilities
- Location

Direct



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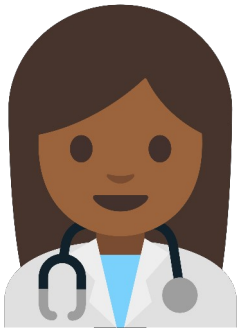
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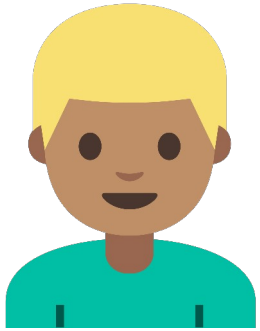


What do the stakeholders **value**?



Direct (Doctor)

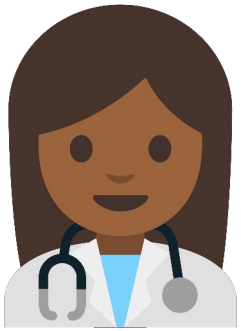
- **Accuracy**
- Training/skill set
- Ease of use
- Research advances



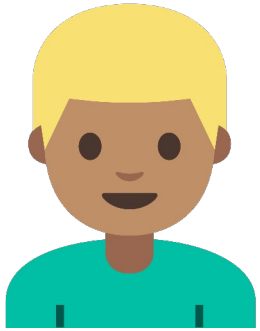
Indirect (Patient)

- Personal care
- **Being informed / autonomy**
- **Trust**
- Privacy

Do **value tensions** arise?



Direct (Doctor)

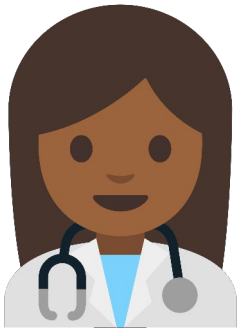


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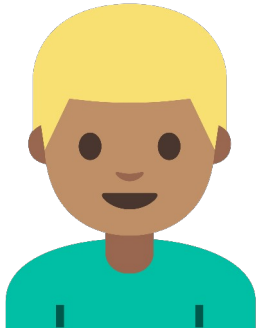
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Indirect (Patient)

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How can we **resolve** value tensions?

1

Is one of the conflicting values considered a **right**?

2

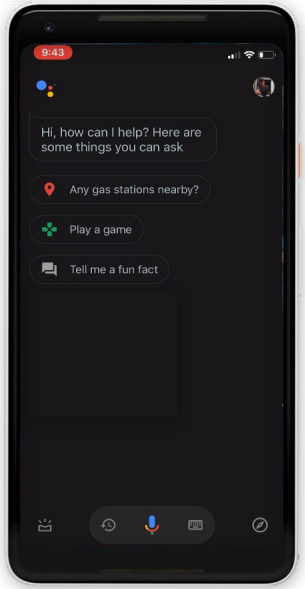
Do the stakeholders **prioritize** certain values?

3

Does the value have a **threshold** for sufficiency?

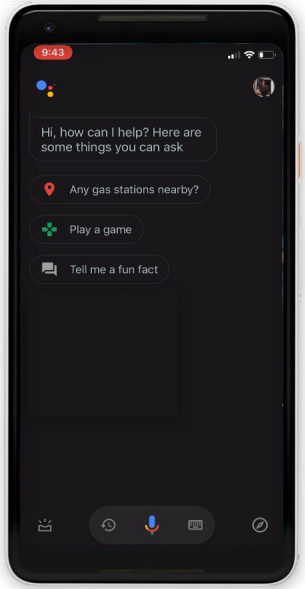
Consider diverse perspectives

How does this technology look when analyzed from **diverse perspectives**, particularly those that are historically marginalized or under-represented?



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Example:
How would the choice of command phrases look to...

- People who are **non-native English speakers**?
- People who speak **African American Vernacular English**?
- People with **speech impairments**?

Consider social context

What is the **social context** in which the technology will be used?
What features of this context are likely to impact **access, adoption and use**?



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Example:
If our intended users span different countries...

- Will there be a **digital divide**?
- Do the countries have different **rules** and **regulations** for **data protection**?
- Do **customs** impact how users **prioritize values**?

We identified **who** we are building for…

But ***how*** are they going to use it?

How are they going to use it?

Multistability

Even if we design technology for a specific purpose, it may be compatible with immoral, malicious or simply unintended purposes

Google pledges to hold off on selling facial recognition technology

The company says important policy questions need to be answered first.

New AI fake text generator may be too dangerous to release, say creators

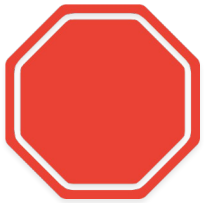
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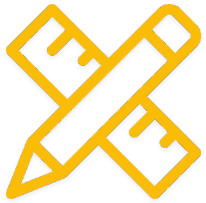
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Non-Intended Use—What should we do?



STOP

Stop developing the technology altogether



CHANGE

Implement design choices to prevent or mitigate non-intended use



REPURPOSE

Embrace non-intended use and redesign for new purpose